Neanderthal Glue May Have Doubled as Germ-Fighting Medicine

From: ScienceDaily Top Environment Mined: 2026-10-12 5 min read

Neanderthals were ancient humans who lived in Europe and parts of Asia long ago. They made a sticky substance called birch tar by heating the bark of birch trees. **Archaeologists** have found traces of it at many Neanderthal sites, often still stuck to stone tools. For decades, scientists thought **Neanderthals** used birch tar mainly as glue, to attach stone blades and points to wooden handles.

A new study suggests birch tar may have had a second job: helping protect wounds from **infection**.

Researchers from the University of Cologne in Germany, the University of Oxford in England, the University of Liège in Belgium and Cape Breton University in Canada did the study. It was published in the science journal PLOS One.

The team made its own birch tar using kinds of trees that grew during Neanderthal times. Instead of modern factory methods, they copied techniques that **archaeologists** think **Neanderthals** used. In one method, they sealed birch bark in a pit underground and heated it with little or no air. In a simpler method, they burned bark next to a stone, and tar collected on the cooler stone surface.

Then they tested the tar against **bacteria**. Every sample they made was able to slow the growth of Staphylococcus aureus, a type of **bacteria** that often causes wound **infections**. Some kinds of this germ have become hard to treat with modern **antibiotics**.

The idea came partly from other clues. Traditional societies around the world have used birch tar as medicine, and there is growing evidence that **Neanderthals** used plants and cared for one another.

"There is also growing evidence of **medicinal** practices and the use of plants among **Neanderthals**," said Tjaark Siemssen, who led the study. The results suggest **Neanderthals** may have had a useful natural medicine.

Word Treasure

Neanderthals -- Ancient humans who lived in Europe and parts of Asia long ago.
Archaeologists -- Scientists who study the past by digging up and examining old objects.
bacteria -- Tiny living things with one cell; some cause infections.
infection -- Illness or harm caused when germs get into the body, such as in a cut.
antibiotics -- Medicines that kill bacteria or stop them from growing.
medicinal -- Used to treat illness or help healing.

Background you need

Neanderthals were close relatives of modern humans who made stone tools and lived through ice age times.

Birch tar is made by heating birch bark, and it works like a strong natural glue.

Archaeologists learn about ancient people by studying tools and leftovers, and scientists sometimes recreate old methods to test ideas.

Some bacteria can infect wounds, and today doctors often treat those infections with antibiotics.

Break it down

LEAD: Neanderthal birch tar may have been medicine as well as glue
KEY EVENT: Researchers recreated ancient birch tar and tested it on bacteria
+- WHO: Teams from Cologne, Oxford, Liège and Cape Breton universities
+- HOW: Underground pit heating and burning bark next to a stone
EVIDENCE: Every sample slowed Staphylococcus aureus, a wound-infecting germ
SUPPORT: Traditional medicinal use and signs Neanderthals used plants and cared for others
OPEN QUESTION: Did Neanderthals actually use birch tar to treat wounds?

Why it matters

The study suggests Neanderthals may have understood more about healing than once thought. It also shows how experiments today can test ideas about life long ago.

Test what you remember

1. How did Neanderthals make birch tar?

- (A) By heating birch bark
- (B) By boiling seawater
- (C) By crushing bones
- (D) By melting rocks

2. What did scientists long think birch tar was mainly used for?

- (A) Fuel for fires
- (B) Food
- (C) Paint for caves
- (D) Glue to attach stone blades to handles

3. Which journal published the study?

- (A) Nature
- (B) Science Advances
- (C) Small
- (D) PLOS One

4. What happened when the tar was tested against bacteria?

- (A) The bacteria grew faster
- (B) Only one sample worked
- (C) Every sample slowed Staphylococcus aureus
- (D) The tests could not be finished

5. Which is one method the team used to make tar?

- (A) Freezing bark overnight
- (B) Sealing bark in an underground pit and heating it
- (C) Using a modern factory machine
- (D) Mixing bark with sugar

6. Who led the study?

- (A) Tjaark Siemssen
- (B) Krasimir Vasilev
- (C) Noelle Bryan
- (D) Kostya Trachenko

Answer key (try the quiz first!): 1: A 2: D 3: D 4: C 5: B 6: A

Questions to think about

1. Tjaark Siemssen, study leader: Says there is growing evidence of medicinal practices and plant use among Neanderthals.

MY THOUGHTS (at least 20 words):